

Ensemble Eval Experiments

Flow-Factory

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1 Experimental Setup

Offline **ensemble-eval** on Stable Diffusion 3.5 Medium: multiple FlowGRPO LoRA checkpoints are fused at each denoising step, then evaluated on three held-out test sets (**geneval**, **pickscore**, **ocr**). Checkpoints: `jieliu/SD3.5M-FlowGRPO-Text` (OCR), `jieliu/SD3.5M-FlowGRPO-PickScore`, `jieliu/SD3.5M-FlowGRPO-OCR`. Rows with $n \geq 2$ teachers use uniform checkpoint averaging (AVG) unless marked PCGRAD. Table 1: **bold** = single-teacher in-domain metric; gray = in-domain for multi-teacher rows; red/blue footnotes = change vs. the best single-teacher baseline among teachers in that row (max over `1_geneval`, `1_ocr`, `1_pickscore` present in the ensemble). Table 2 gray cells mark in-domain metrics (appendix).

2 Main Results

Table 1: Cross-benchmark eval metrics (mean over test set). **Bold**: single-teacher in-domain. Gray + red/blue footnotes: multi-teacher vs. max single-teacher in row.

Teachers	Agg.	GenEval benchmark		PickScore benchmark	OCR benchmark	
		GenEval	PickScore	PickScore	OCR	PickScore
SD3.5 (base)	—	0.657	0.875	0.835	0.574	0.866
geneval	—	0.937	0.888	0.829	0.656	0.862
ocr	—	0.685	0.876	0.837	0.931	0.869
pickscore	—	0.641	0.918	0.895	0.690	0.915
geneval+ocr	AVG	0.850-0.087	0.888-0.000	0.837+0.000	0.875-0.055	0.874+0.005
geneval+pickscore	AVG	0.830-0.107	0.908-0.010	0.873-0.021	0.675-0.015	0.899-0.017
ocr+pickscore	AVG	0.680-0.005	0.901-0.017	0.873-0.022	0.865-0.066	0.899-0.016
geneval+ocr+pickscore	PCGRAD	0.794-0.143	0.901-0.017	0.864-0.031	0.844-0.087	0.894-0.022

3 Full Eval Metrics

Table 2: All scalar eval/ metrics (std and GenEval per-tag breakdowns).

Teachers	Agg.	GenEval benchmark														PickScore benchmark		OCR benchmark	
		GenEval						PickScore						PickScore		OCR	PickScore		
		Std	Color attr	Colors	Counting	Position	Single obj	Two obj	Std	Color attr	Colors	Counting	Position	Single obj	Two obj	Std	Std		
SD3.5 (base)	—	0.395	0.521	0.711	0.761	0.450	0.844	0.636	0.042	0.900	0.879	0.875	0.866	0.855	0.873	0.057	0.393	0.039	
geneval	—	0.205	0.854	0.867	0.994	0.950	1.000	0.982	0.038	0.920	0.890	0.890	0.869	0.864	0.887	0.056	0.346	0.038	
ocr	—	0.375	0.531	0.689	0.802	0.490	0.867	0.664	0.042	0.902	0.882	0.879	0.874	0.856	0.874	0.057	0.173	0.041	
pickscore	—	0.407	0.479	0.622	0.913	0.440	0.800	0.655	0.043	0.949	0.919	0.917	0.926	0.881	0.921	0.058	0.350	0.039	
geneval+ocr	Avg	0.307	0.719	0.800	0.898	0.730	0.911	0.945	0.040	0.921	0.886	0.889	0.880	0.867	0.884	0.056	0.231	0.040	
geneval+pickscore	Avg	0.319	0.740	0.800	0.904	0.770	0.933	0.873	0.041	0.947	0.907	0.907	0.906	0.875	0.909	0.057	0.349	0.038	
ocr+pickscore	Avg	0.387	0.490	0.711	0.867	0.550	0.778	0.682	0.042	0.935	0.897	0.903	0.905	0.869	0.900	0.057	0.253	0.040	
geneval+ocr+pickscore	PCGRAD	0.338	0.625	0.844	0.959	0.650	0.867	0.818	0.041	0.940	0.899	0.898	0.898	0.870	0.900	0.058	0.259	0.039	